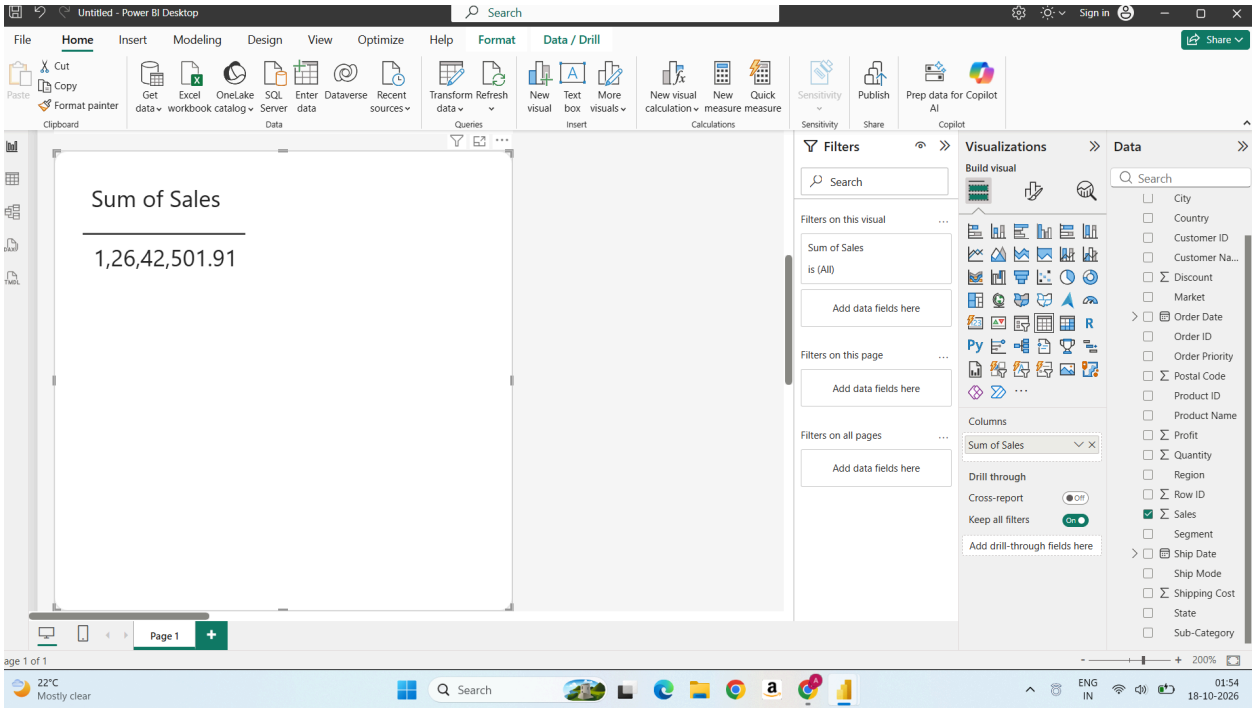
Power BI Visuals

Question 1: Create a Card Visual to show Total Sales.

- **Visual Type:** Card (123 icon)
- **Fields Used:** Sales (Sum of Sales)
- **Steps Performed:**
    1. Selected the **Card** visual icon from the Visualizations pane.
    2. Dragged the **Sales** field into the **Value** bucket.
- **Result / Output:** Total Sales = **$12,642,501.91** (~$12.64M).

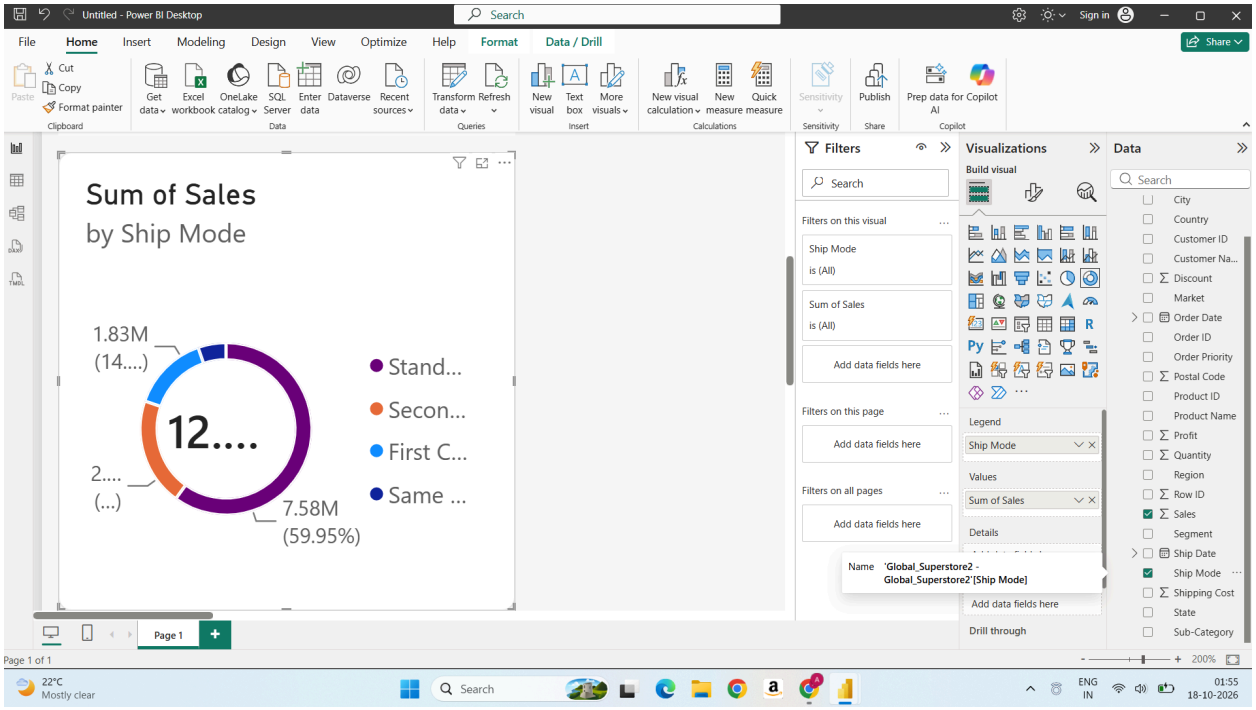


Question 2: Create a Donut Chart showing Total Sales by Ship Mode and which ship mode generates the highest sales?

- **Visual Type:** Donut Chart
- **Fields Used:**
    - **Legend:** Ship Mode
    - **Values:** Sales
- **Steps Performed:**

- Selected the **Donut Chart** visual from the Visualizations pane.
- Dragged **Ship Mode** into the **Legend** well.
- Dragged **Sales** into the **Values** well.

- **Findings / Analysis:**
  - **Standard Class** generates the highest sales at **$7,578,652.00** (~60% of total sales).
  - **Second Class** comes second (~$2.57M), followed by **First Class** (~$1.83M) and **Same Day** (~$667K).

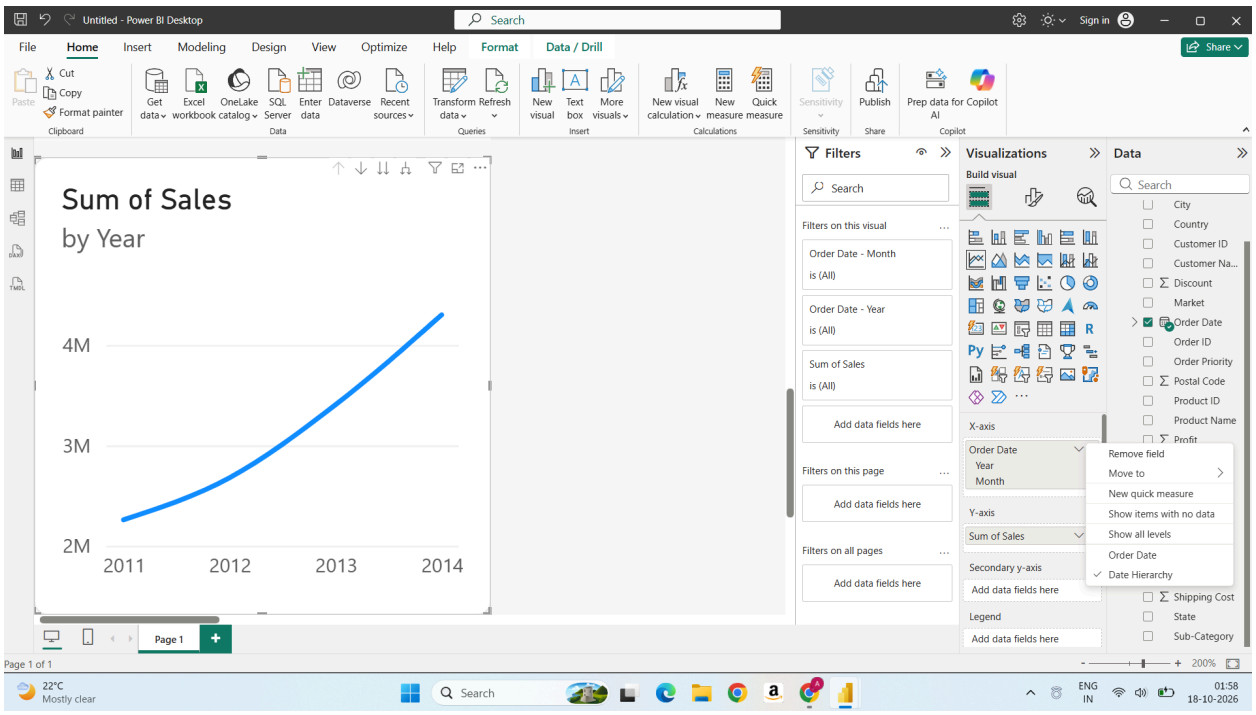


Question 3: Create a Line Chart showing the Sales Trend over time (by Month and Year). Is there any noticeable seasonal pattern in sales?

- **Visual Type:** Line Chart
- **Fields Used:**
  - **X-axis:** Order Date (Hierarchy: Year and Month)
  - **Y-axis:** Sales

- **Steps Performed:**
  - Selected the **Line Chart** visual from the Visualizations pane.
  - Dragged **Order Date** into the **X-axis** field well and set granularity to Year and Month.
  - Dragged **Sales** into the **Y-axis** field well.
- **Findings / Analysis:**
  - **Seasonal Pattern Identified:** Yes, there is a strong recurring seasonal trend. Sales consistently peak during the Q4 months (**November and December**) each

year due to holiday shopping and year-end inventory clearances. Conversely, Q1 (**January and February**) systematically experiences the lowest sales volume.

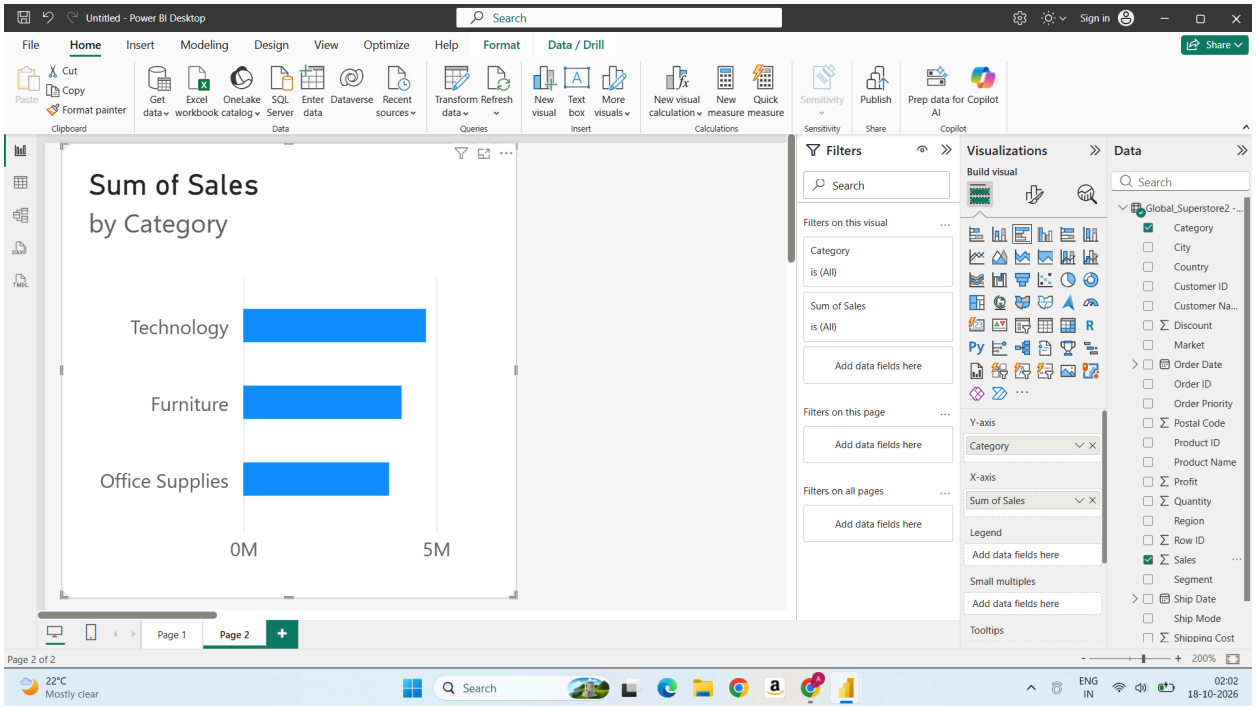


Question 4: Create a Bar Chart to compare Sales by Category.

- **Visual Type:** Clustered Bar Chart (or Column Chart)
- **Fields Used:**
    1. **Y-axis / Axis:** Category

    2. **X-axis / Values:** Sales

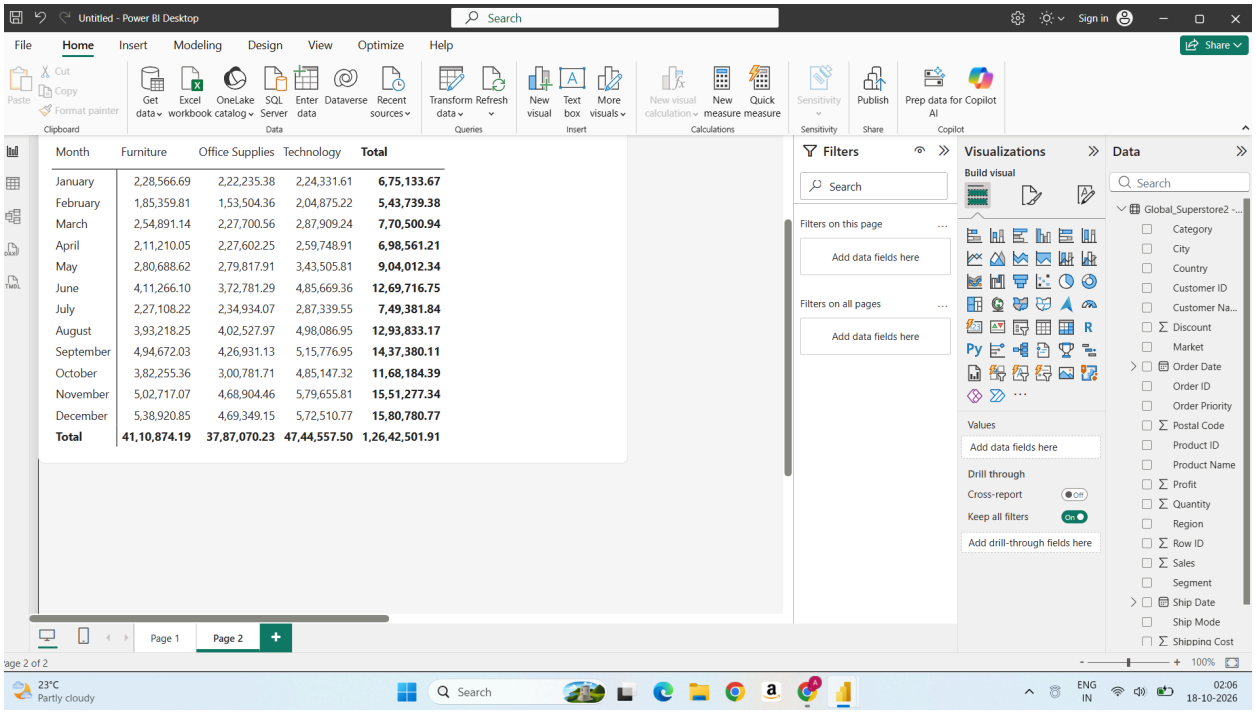
- **Steps Performed:**
    1. Selected the **Clustered Bar Chart** visual from the Visualizations pane.
    2. Dragged **Category** into the Y-axis / Legend field.
    3. Dragged **Sales** into the X-axis / Values field.
- **Findings / Analysis:**
    1. **Technology:** Highest Sales at **$4,744,557.50**
    2. **Furniture:** Second Highest at **$4,110,874.19**
    3. **Office Supplies:** Third at **$3,787,070.23**



Question 5: Create a Matrix Visual to display Sales by Month and Category and identify the month with the highest sales.

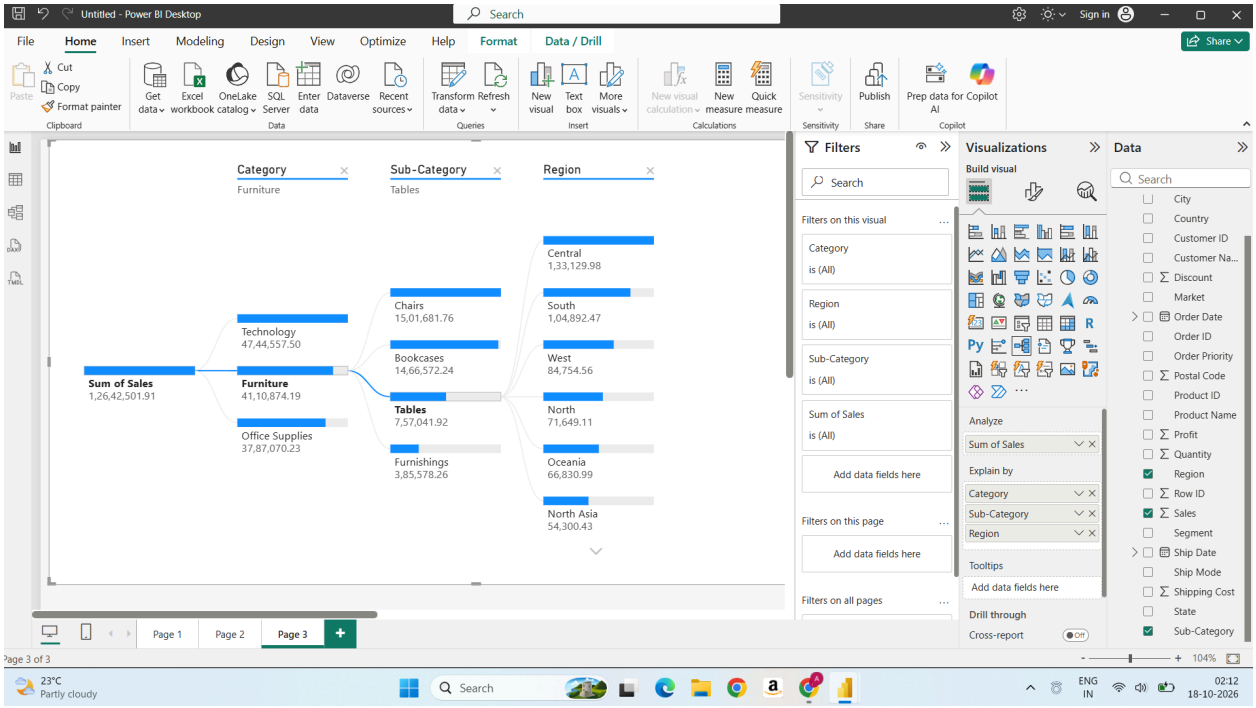
- **Visual Type:** Matrix
- **Fields Used:**
  - **Rows:** Order Date (Month / Year)
  - **Columns:** Category
  - **Values:** Sales
- **Steps Performed:**
  - Selected the **Matrix** visual from the Visualizations pane.
  - Placed **Order Date (Month)** in the **Rows** section.
  - Placed **Category** in the **Columns** section.
  - Placed **Sales** in the **Values** section.

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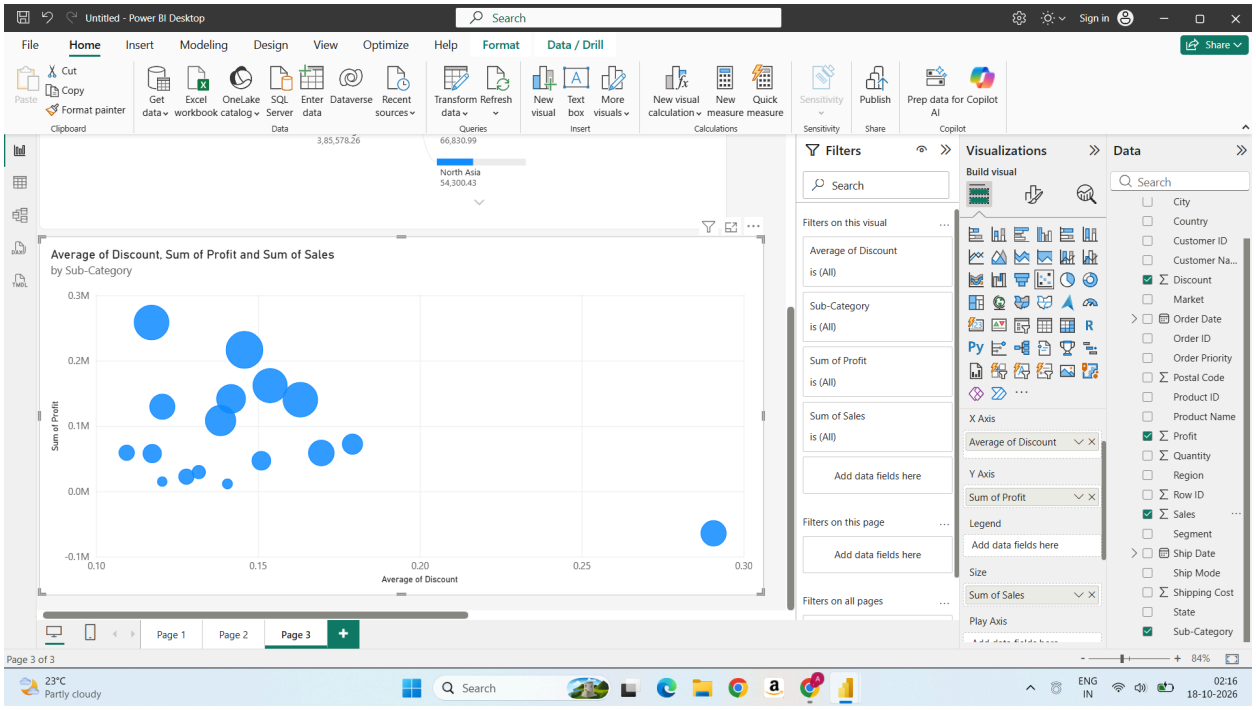
Question 6: Create a Decomposition Tree for Sales Breakdown.

- **Visual Type:** Decomposition Tree
- **Fields Used:**
  - **Analyze:** Sales
  - **Explain By:** Category, Sub-Category, Region
- **Steps Performed:**
  - Selected the **Decomposition Tree** visual from the Visualizations pane.
  - Dragged **Sales** into the **Analyze** bucket.
  - Dragged **Category**, **Sub-Category**, and **Region** into the **Explain By** bucket.
  - Expanded nodes interactively starting from Total Sales down to Category, Sub-Category, and Region.
- **Findings / Analysis:**
  - Allows root-cause path analysis showing how top-level sales ($12.64M) break down iteratively across business categories, sub-categories, and geographic regions.



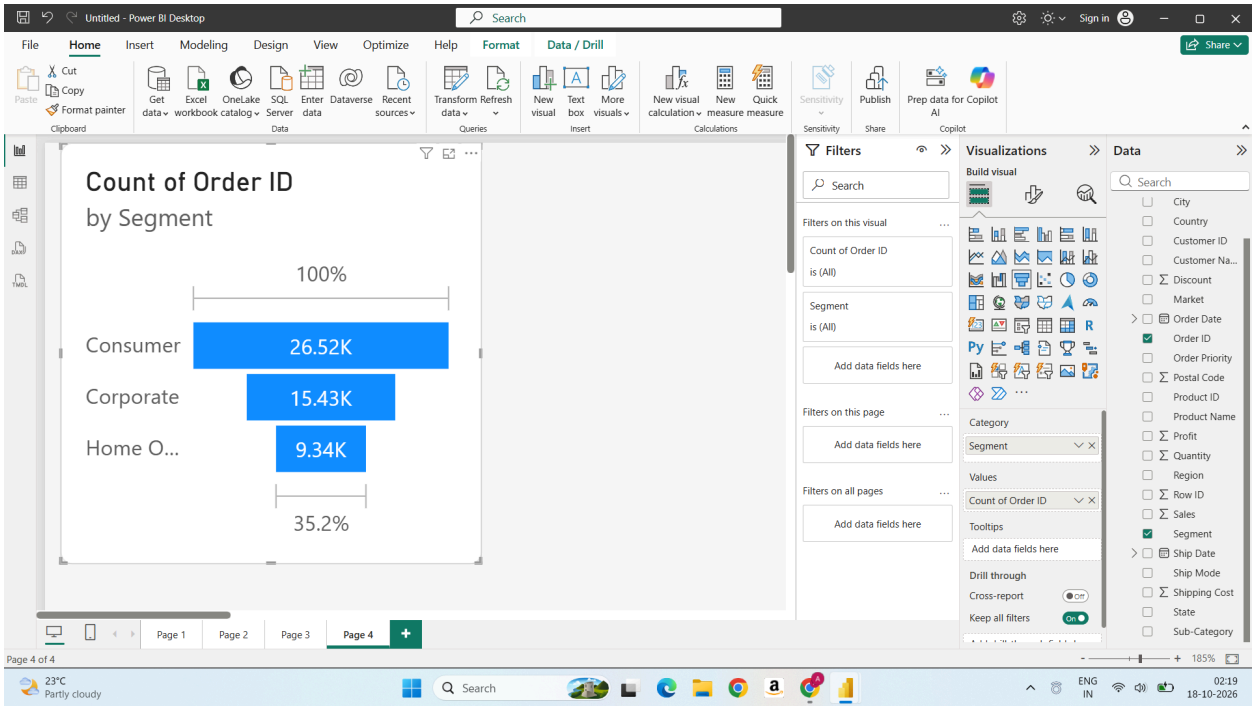
Question 7: Create a Scatter Chart to compare Discount vs. Profit by Sub-Category. Are there any outliers where high discounts resulted in negative profit?

- **Visual Type:** Scatter Chart
- **Fields Used:**
    - **Values / Details:** Sub-Category
    - **X-axis:** Discount (Average Discount)
    - **Y-axis:** Profit (Sum of Profit)
    - **Size:** Sales
- **Steps Performed:**
    - Selected the **Scatter Chart** visual from the Visualizations pane.
    - Placed **Sub-Category** in the **Values / Details** well.
    - Dragged **Discount** (set to Average) to the **X-axis**.
    - Dragged **Profit** (Sum) to the **Y-axis**.
- **Findings / Analysis:**
    - **Outlier Identified: Tables** is a significant negative outlier. With an average discount of **~29.1%** (highest among all sub-categories), it resulted in a cumulative negative profit loss of **-$64,083.39**. Higher discount rates directly eroded profitability in this category.



Question 8: Use a Funnel Chart to show the number of orders by Customer Segment.

- **Visual Type:** Funnel Chart
- **Fields Used:**
  - **Category:** Segment
  - **Values:** Order ID (Count / Distinct Count)
- **Steps Performed:**
  - Selected the **Funnel Chart** visual from the Visualizations pane.
  - Dragged **Segment** into the **Category** well.
  - Dragged **Order ID** (Count / Count Distinct) into the **Values** well.
- **Findings / Analysis:**
  - **Consumer:** 26,518 order rows (~51.7% of total volume)
  - **Corporate:** 15,429 order rows (~30.1% of total volume)
  - **Home Office:** 9,343 order rows (~18.2% of total volume)

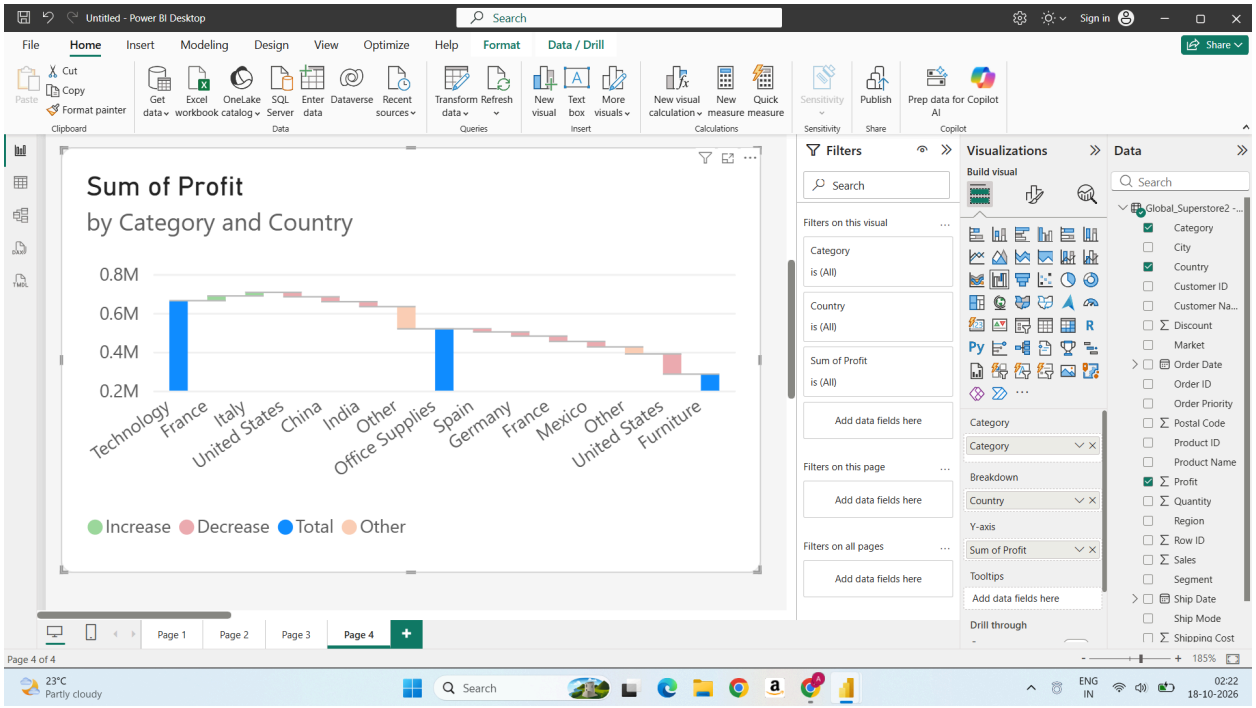


Question 9: Use a Waterfall Chart to analyze how Profit changes by Category within the countries.

- **Visual Type:** Waterfall Chart
- **Fields Used:**
    - **Category:** Category

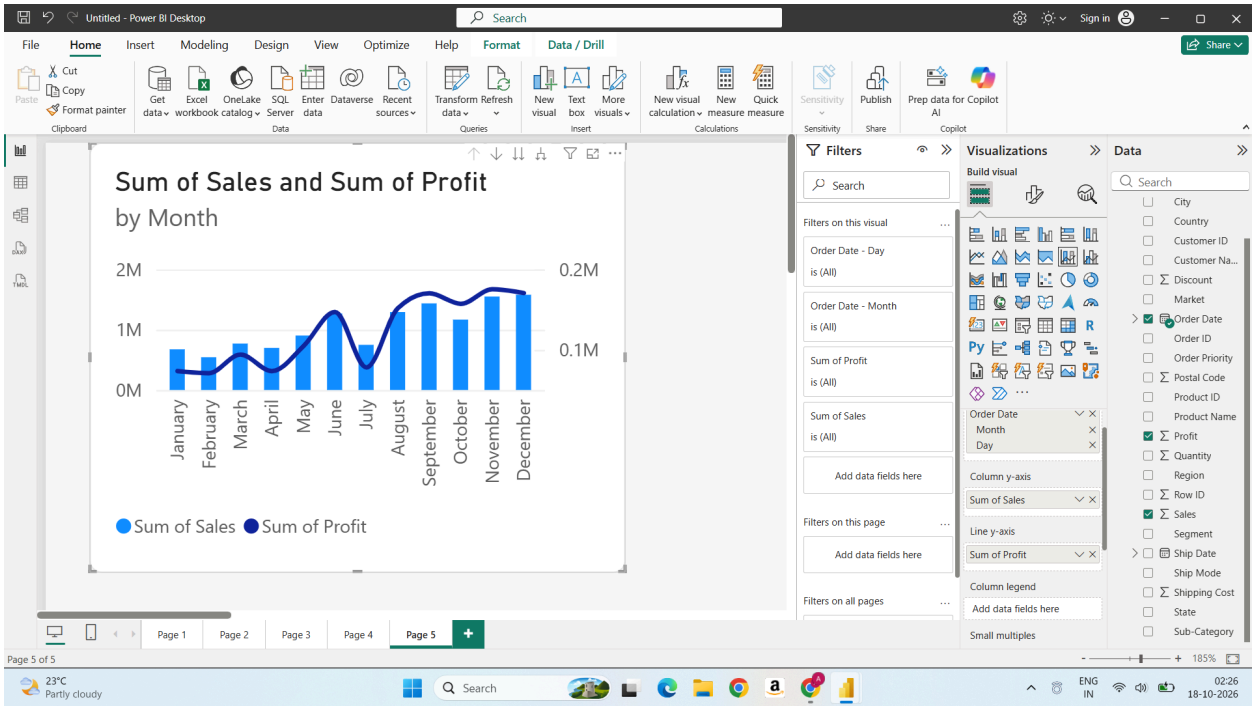
    - **Breakdown / Sub-category:** Country (or Region)
    - **Y-axis / Values:** Profit

- **Steps Performed:**
    - Selected the **Waterfall Chart** visual from the Visualizations pane.
    - Placed **Category** in the **Category** bucket.
    - Placed **Profit** in the **Y-axis / Values** bucket.
    - Added **Country** (or Region) to **Breakdown** for regional profit variance analysis.
- **Findings / Analysis:**
    - Illustrates cumulative building of profit: **Technology** adds the largest positive profit gain (+$663,778.73), followed by **Office Supplies** (+$518,473.83) and **Furniture** (+$285,204.72).



Question 10: Create a Dual-Axis Line Chart to show Total Sales and Profit Over Time (by Month).

- **Visual Type:** Line and Stacked Column Chart / Line Chart with Secondary Y-Axis
- **Fields Used:**
    - **X-axis:** Order Date (Month & Year)
    - **Y-axis / Column Values:** Sales
    - **Secondary Y-axis / Line Values:** Profit
- **Steps Performed:**
    - Selected **Line and Stacked Column Chart** (or Line Chart with Secondary Y-Axis) from the Visualizations pane.
    - Dragged **Order Date** (Month) into the **X-axis**.
    - Dragged **Sales** into the **Column values / Y-axis**.
    - Dragged **Profit** into the **Line values / Secondary Y-axis**.
- **Findings / Analysis:**
    - Shows a strong positive correlation between Sales volume and Profitability over time. As sales spike during Q4 (Nov-Dec), profit trends upward proportionally, though profit margins vary depending on discount application in given months.


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Modeling
Design
View
Optimize
Help
Format
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Share
Sum of Sales and Sum of Profit
by Month
2M
1M
0M
0.2M
0.1M
January
February
March
April
May
June
July
August
September
October
November
December
Sum of Sales
Sum of Profit
Filters
Search
Filters on this visual
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Sum of Sales
Line y-axis
Sum of Profit
Column legend
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City
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Customer ID
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Market
Order Date
Order ID
Order Priority
Postal Code
Product ID
Product Name
Profit
Quantity
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